

VI. Regression Test Suite

Library Management System

Test Team

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Note: This regression test suite is designed to verify that recent code changes have not adversely affected existing core functionalities. It focuses on the primary "happy path" scenarios and critical features identified in the SRS. These tests should be executed after every bug fix or feature implementation.

VI. Regression Test Cases

Table 1: Regression Test Suite

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: Authentication & Core Access (Smoke Tests)			
TC_REG.AUTH.001	Successful login with valid Librarian credentials	- Navigate to the login page. - Enter valid Librarian email and password. - Click the 'Login' button.	- User is authenticated successfully. - User is redirected to the Librarian dashboard. - Librarian-specific functionalities (e.g., 'User Management') are visible.
TC_REG.AUTH.002	Successful login with valid Member credentials	- Navigate to the login page. - Enter valid Member email and password. - Click the 'Login' button.	- User is authenticated successfully. - User is redirected to the books page. - Librarian-specific functionalities are not visible.
TC_REG.AUTH.003	Verify Member cannot access Librarian-specific URLs	- Login as a Member. - Attempt to navigate directly to the User Management URL (e.g., /users).	- Access is denied. - User is redirected to an error page or the member dashboard.
TC_REG.AUTH.004	Successful logout for a logged-in user	- Login as any user (Librarian or Member). - Click the 'Logout' button/link. - Attempt to use the browser's "back" button to access a protected page.	- User is logged out successfully and redirected to the login page. - The user session is terminated, and access to protected pages is denied.
Sub-Section: Core CRUD Functionality (Librarian)			
TC_REG.CRUD.001	Librarian can add a new book	- Login as Librarian. - Navigate to Book Management (/books). - Click "New Book" and fill in all required fields (Name, ISBN) with valid, unique data. - Submit the form.	- A success message is displayed. - The new book record is created and appears in the book list.
TC_REG.CRUD.002	Librarian can view and update an existing book	- Login as Librarian and navigate to Book Management. - Select an existing book and choose to edit it. - Change a field (e.g., update the summary). - Save the changes.	- The book details are displayed correctly in the edit form. - A success message is displayed after saving. - The book list or detail view reflects the updated information.
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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_REG.CRUD.003	Librarian can register a new user	• Login as Librarian. • Navigate to User Management (/users). • Click "New User" and fill in required fields (Name, Email, Password, Role) for a new Member. • Submit the form.	• A success message is displayed. • The new user appears in the user list. • The newly created user can log in successfully.
Sub-Section: Core Use Cases / User Workflows			
TC_REG.FLOW.001	Member can borrow an available book	• Precondition: A book exists and is marked as available. • Login as Member. • Navigate to Borrowal Management (/borrowals). • Click "New Borrowal". • Select the available book. • Submit the form.	• A new borrowal record is created successfully with 'Borrowed' status. • A success message is displayed. • The availability status of the book is updated to 'false' (unavailable).
TC_REG.FLOW.002	Member can view their own borrowal history	• Precondition: A member has at least one active or past borrowal record. • Login as the Member. • Navigate to the Borrowal Management page (/borrowals).	• The page displays a list of borrowal records. • The list contains ONLY the borrowals belonging to the logged-in member.
TC_REG.FLOW.003	Librarian can update a borrowal status (e.g., return a book)	• Precondition: A book has been borrowed by a member. • Login as Librarian. • Navigate to Borrowal Management. • Find the active borrowal record and edit it. • Change the status from 'Borrowed' to 'Returned'. • Save the changes.	• The borrowal record's status is successfully updated to 'Returned'. • The corresponding book's availability status is updated back to 'true' (available).